

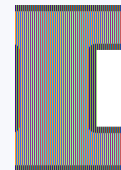


FABRIC Capabilities and Use-cases

I. Baldin, UNC Chapel Hill/RENCI

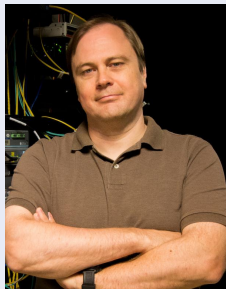
NDNcomm

Sep 10, 2020



FABRIC Leadership Team

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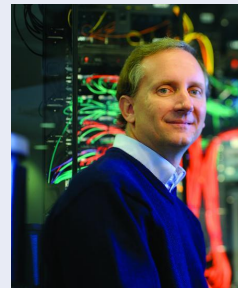
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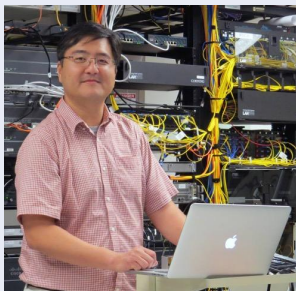
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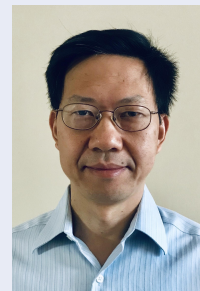
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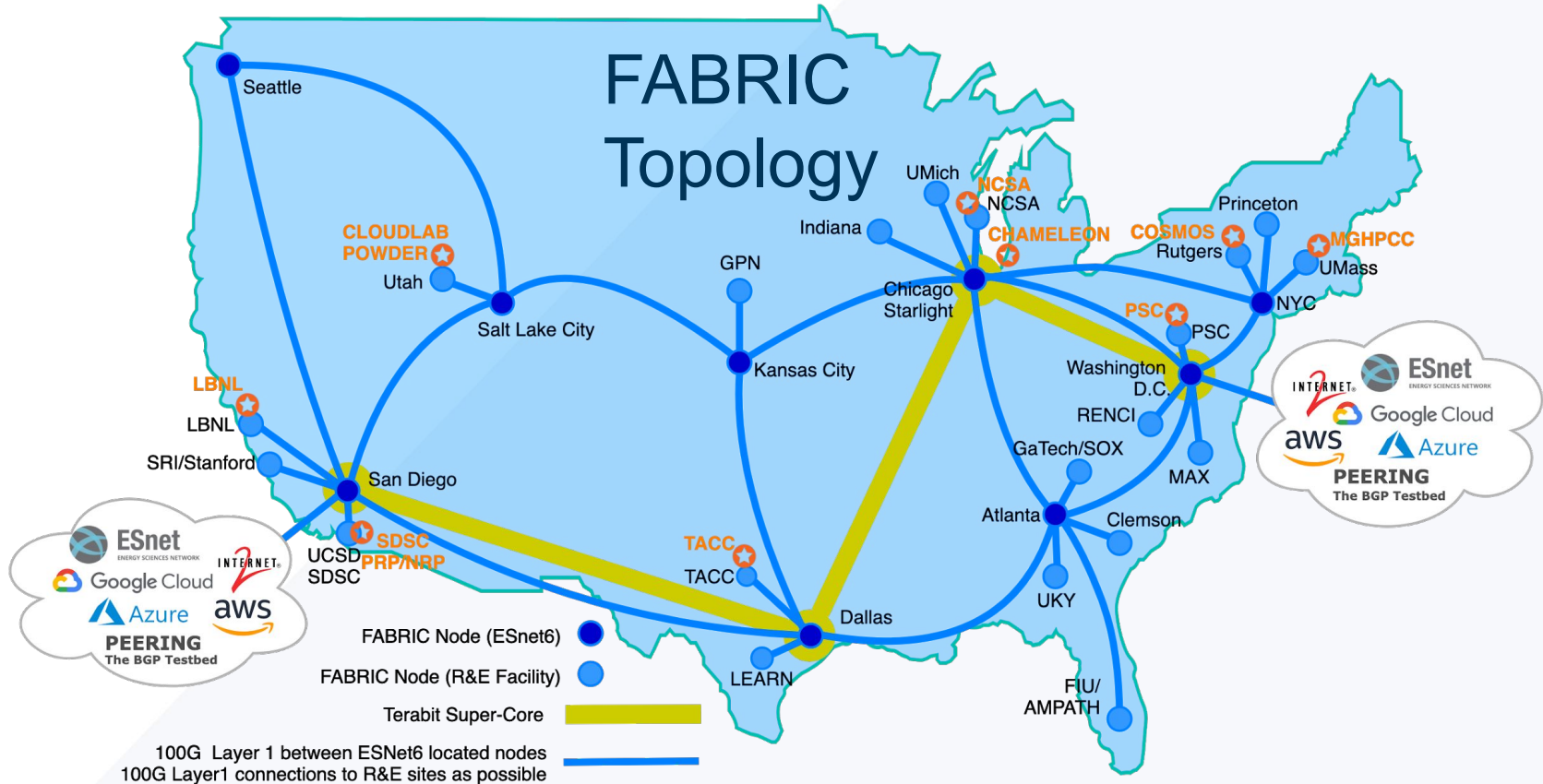
What is Fabric?

FABRIC enables a completely *new paradigm for distributed protocols and applications*

- A nation-wide programmable network with significant compute and storage at each node, allowing users to run computationally intensive programs throughout the network.
- Provides GPUs, FPGAs, and network processors (NICs) inside the network.
- Supports quality of service (QoS) using dedicated optical 100G links or dedicated L2
- Interconnects national facilities: HPC centers, cloud & wireless testbeds, commercial clouds, the Internet, and edge nodes at universities and labs.
- Allows you to design and test programs that run at any node in the network, not just the edge or cloud.



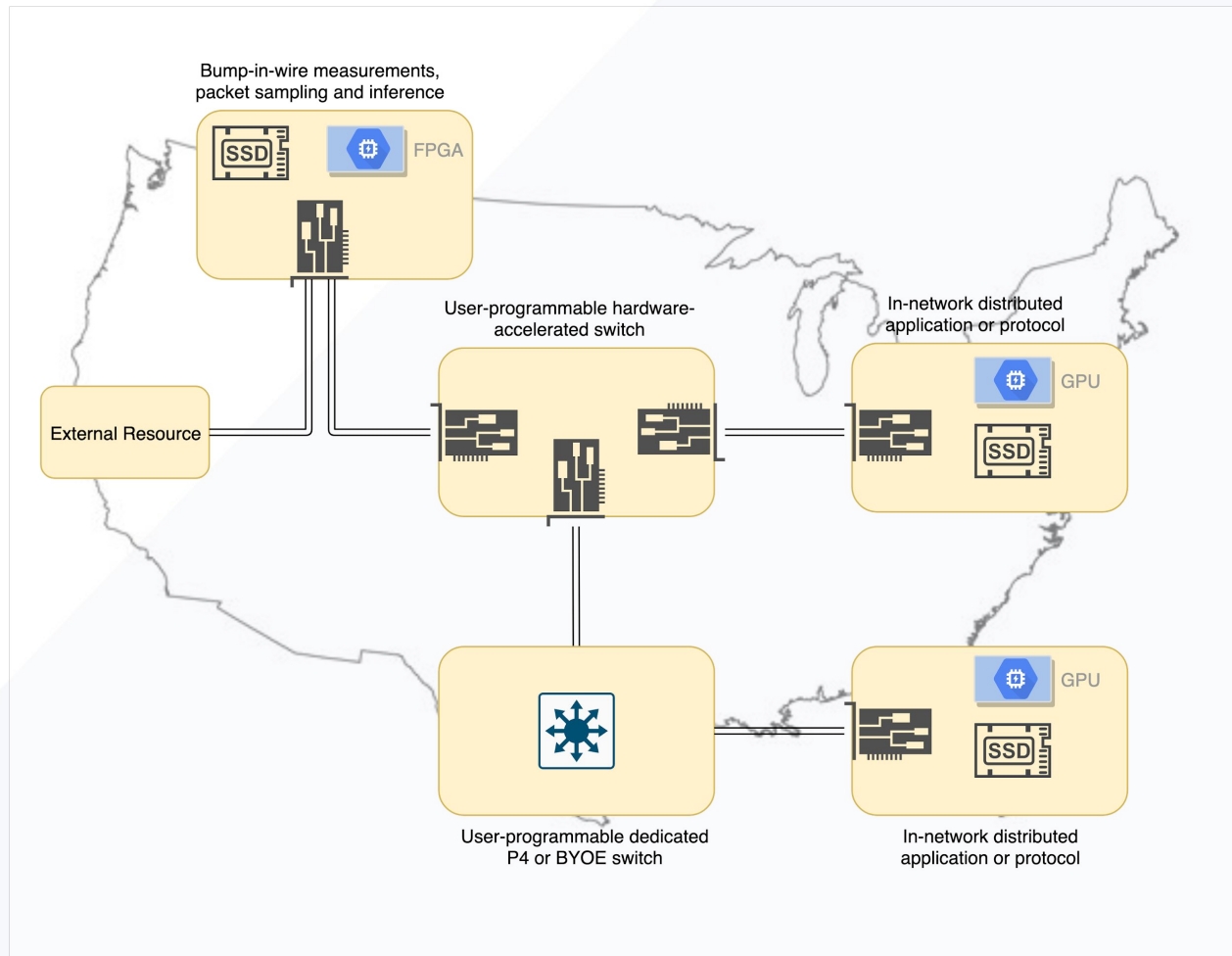
FABRIC Topology



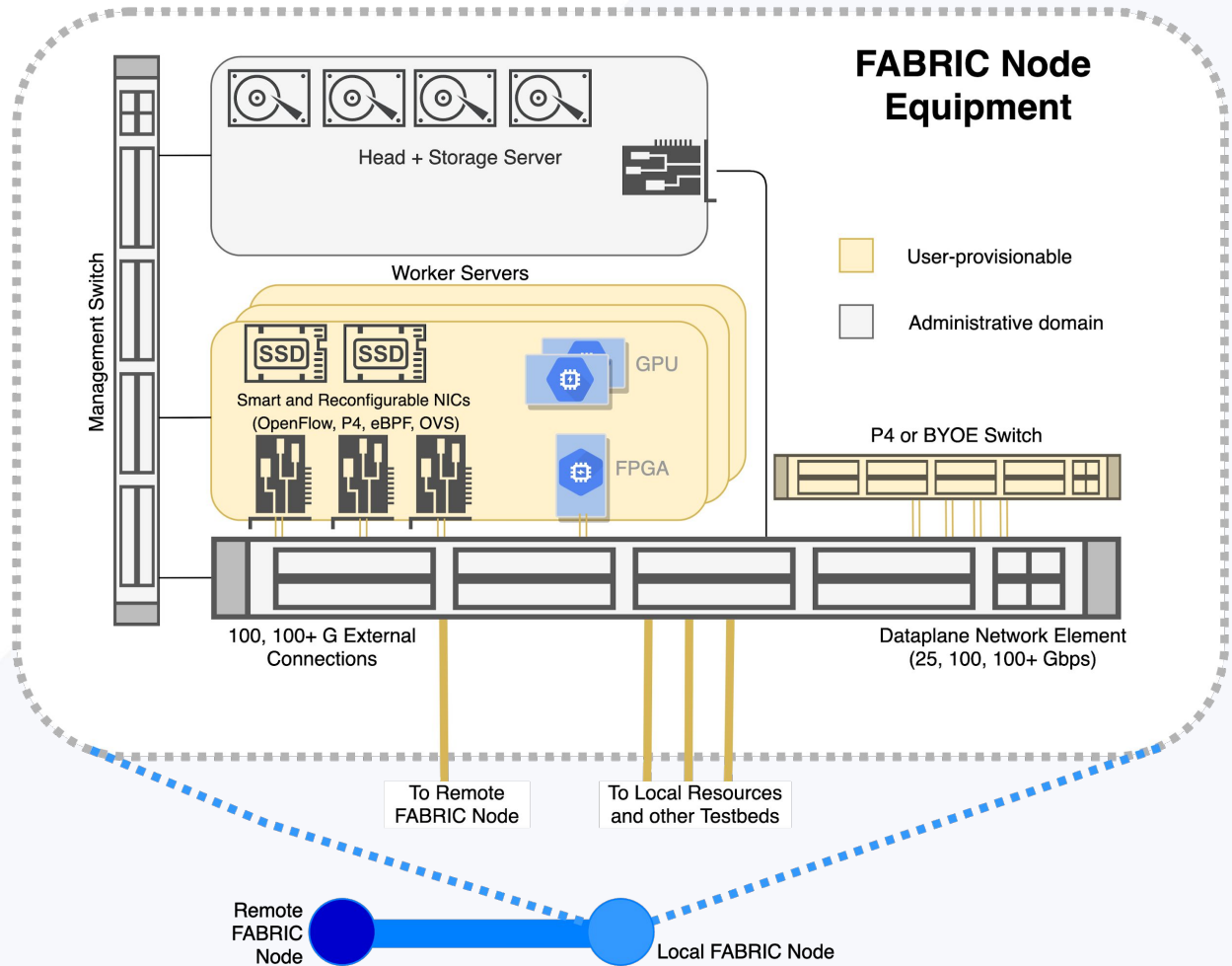
FABRIC Nodes

- Interpose compute and storage into the path of fast packet flows
- Rack of high-performance servers (Dell 7525) with:
 - 32-core AMD 7532 with 512G RAM
 - GPUs (RTX 6000 and T4), FPGA compute accelerators
 - Storage - experimenter provisionable 1TB NVMe drives in servers and a pool of ~250TB rotating storage at each site.
 - Network ports connect to a 100G+ switch, programmable through control software
- Reconfigurable Network Interface Cards
 - FPGAs (P4)
 - Mellanox ConnectX-5 and ConnectX-6
 - Multiple interface speeds (25G, 100G, 200G+(future))
- Kernel Bypass/Hardware Offload
 - VM/Containers sized to support full-rate DPDK for access to Programmable NICs, FPGA, and GPU resources

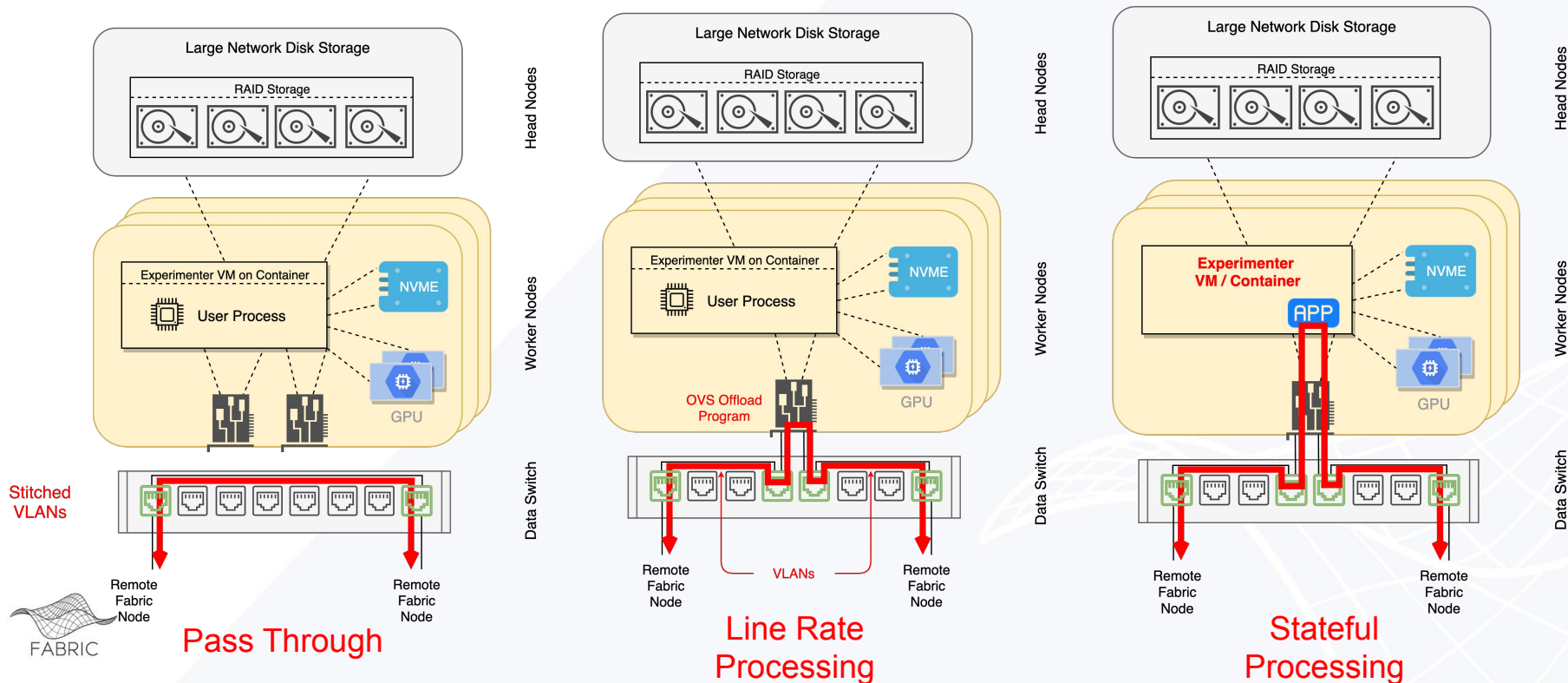
Conceptual FABRIC Experiment



Conceptual FABRIC Node



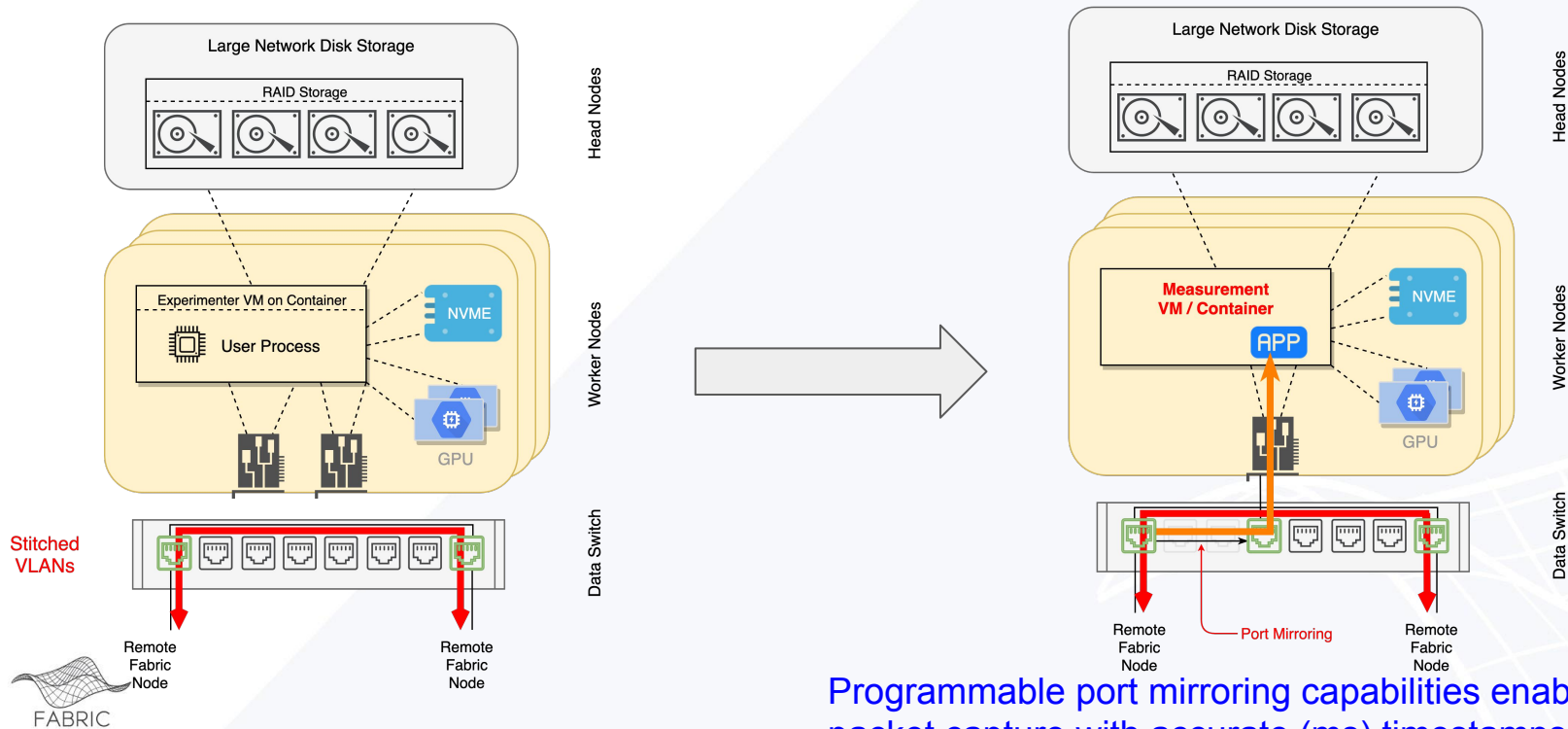
FABRIC Use Cases



FABRIC Measurement Hardware

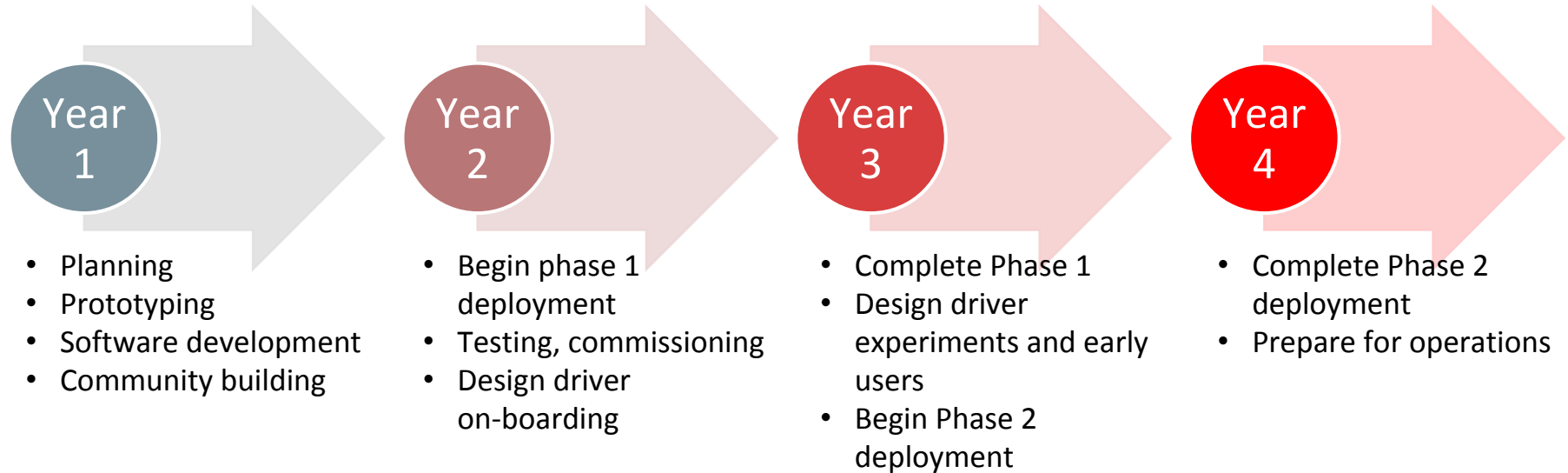
- GPS-disciplined clock source at most sites
 - Fine-grained Precise Measurements
 - Leverages a highly-accurate PTP timing from local GPS receiver
 - Precise timestamping of packets using NIC cards (a.k.a., PacketGPS)
- High-speed NICs capable of accurate packet timestamping
 - Supports high-speed packet capture and (limited) processing
 - Switch supports programmable port mirroring to select flows to capture
- Smart PDUs to measure power
- Optical layer measurements (where available)
- CPU, memory, disk, port/interface utilization and other time-series (software)

FABRIC Use Cases

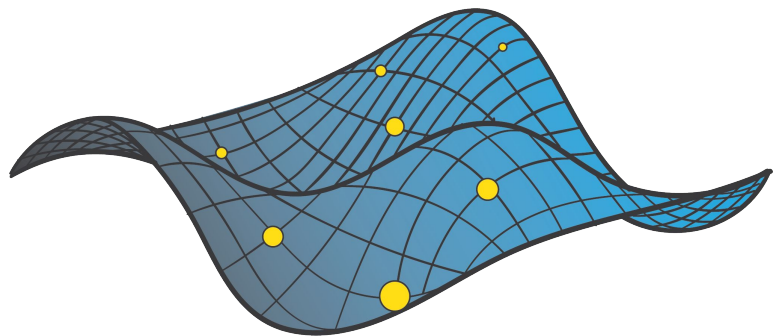


Programmable port mirroring capabilities enable packet capture with accurate (ms) timestamps

Construction Timeline



Thank you!



FABRIC



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<https://whatisfabric.net>